

PKCERT AI & Software Development Internship

Task 28 – Capstone Project: End-to-End AI/Software Application Development — FINAL TASK

Status: Final Task of the Internship

Objective: Design, build, evaluate, and deploy a complete, end-to-end AI-powered or software application that integrates the skills developed across the internship — including prior work on NLP, embeddings, Transformer-based models, and full-stack development. Interns will independently define project scope, prepare data, build and evaluate a model, construct a backend API, build and integrate a frontend, test and deploy the system, and produce professional documentation and a final presentation. This task serves as the culminating deliverable of the internship and is assessed holistically across technical execution, engineering rigor, and communication.

Part A – Capstone Kickoff: Scope & Planning — 10 Marks

- Choose a capstone project idea that meaningfully combines skills acquired during the internship. Suggested directions include an AI-powered chatbot, an image classifier application, a recommendation system, or a sentiment analysis dashboard — an alternative idea may be proposed if it is justified and of comparable scope and difficulty.
- Define a clear problem statement: the target user, the core functionality, and the specific value the application provides.
- Define project scope explicitly, including in-scope and out-of-scope features, to prevent scope creep as the project progresses.
- Select and justify the technical stack for each layer of the system (data/model layer, backend/API layer, frontend/UI layer, deployment target), referencing prior task work where applicable (e.g., Hugging Face/Transformers from Task 27, full-stack patterns from earlier tasks).
- Produce an execution plan for the remaining stages of the project, identifying key milestones, dependencies between stages (e.g., data readiness before model training, model readiness before API integration), and risk areas with mitigation notes.
- Define success criteria and evaluation metrics for the project (e.g., target model accuracy/F1, API latency targets, UI usability expectations) to be revisited in Part D.

Part B – Data Preparation, Model Building & Evaluation — 25 Marks

Data Collection & Preparation

- Identify and justify the data source(s) for the chosen project (public dataset, API-collected data, or synthetically generated data), including licensing/usage considerations.
- Perform data cleaning, preprocessing, and, where applicable, labeling. Document key preprocessing decisions (handling missing values, class imbalance, text normalization/tokenization, feature engineering) and their rationale.
- Split data into training, validation, and test sets, explaining the split strategy and why it is appropriate for the chosen model type and evaluation goals.

Model Building & Training

- Select and justify a modeling approach appropriate to the project (e.g., a fine-tuned/pretrained Transformer model, a classical ML classifier, or a lightweight custom architecture), referencing trade-offs discussed in Task 27 (static vs. contextual embeddings) where relevant.
- Implement the training pipeline, including data loading, model configuration, training loop or fine-tuning procedure, and checkpointing.

- Report training configuration (hyperparameters, epochs, optimizer, loss function) and training/validation curves.

Model Evaluation & Iteration

- Evaluate the trained model against the metrics defined in Part A using the held-out test set, presenting results in tabular form.
- Conduct at least one iteration cycle (e.g., hyperparameter adjustment, additional data cleaning, architecture change, or feature adjustment) in response to evaluation results, and report the before/after comparison.
- Perform basic error analysis: identify representative failure cases and discuss probable causes.

Part C – Backend & Frontend Development, Integration — 20 Marks

API/Backend Development

- Design and implement a backend API (e.g., Node.js/Express, or a Python framework such as FastAPI/Flask, consistent with the stack chosen in Part A) that serves the trained model and exposes the endpoints required by the application.
- Define the API contract (endpoints, request/response schemas, status codes, error handling) and implement input validation.
- Apply the intern's preferred authentication pattern where the application requires user accounts (registration/login forms, JWT issued in an HTTP-only cookie, protected routes via backend middleware), or explicitly justify why authentication is out of scope for this project.

Frontend/UI Development & Integration

- Design and implement a frontend interface appropriate to the project (e.g., React/Next.js) that allows a user to interact with the underlying model through the backend API.
- Integrate the frontend with the backend API, handling loading states, error states, and displaying model outputs in a clear, user-appropriate format.
- Ensure the UI reflects the scope defined in Part A (no unjustified feature additions or omissions).

Part D – Testing, Deployment & Documentation — 20 Marks

Testing, Bug Fixing & Code Cleanup

- Perform functional testing of the end-to-end system (data → model → API → UI), documenting test cases and outcomes.
- Identify and resolve bugs discovered during testing, and perform a code cleanup pass (removing dead code, improving structure/naming, adding inline documentation where necessary).

Deployment

- Deploy the completed capstone project (backend, frontend, and model-serving component as applicable) to a live or reproducible environment (e.g., a cloud/PaaS provider, containerized deployment via Docker, or an equivalent approach), and provide a working access method (URL or clear local run instructions) for evaluation.
- Document any deployment-specific configuration (environment variables, build steps, containerization details).

Documentation

- Write a comprehensive README covering project overview, setup instructions, environment/dependency requirements, and usage examples.
- Produce an architecture diagram illustrating the full system (data/model layer, backend, frontend, deployment) and how components communicate.

- Ensure documentation is sufficient for a third party to set up and run the project independently.

Part E – Final Presentation & Internship Wrap-up — 25 Marks

Presentation Preparation

- Prepare a final presentation (slide deck) covering the problem statement, approach, technical architecture, key results, challenges encountered, and lessons learned.
- Prepare a live or recorded demo of the working capstone application.

Final Polish

- Use remaining time for last-minute fixes, polish, and rehearsal of the presentation and demo. No new features should be introduced at this stage.

Final Presentation & Internship Wrap-up

- Present the capstone project, including a live demo, to the assessing panel/supervisor.
- Submit the GitHub repository link containing the complete, documented codebase.
- Submit all daily logs compiled across the internship.
- Submit the final internship report summarizing skills gained, tasks completed, and overall reflection on the internship experience.

Assessment Breakdown

Assessment Area	Marks
Part A – Capstone Kickoff, Scope & Planning	10
Part B – Data Preparation, Model Building & Evaluation	25
Part C – Backend & Frontend Development, Integration	20
Part D – Testing, Deployment & Documentation	20
Part E – Final Presentation & Internship Wrap-up	25
Total	100

Note: This is the final task of the PKCERT AI & Software Development Internship. The capstone project must demonstrably integrate skills developed across prior tasks (data/text processing, embeddings and Transformer-based modeling per Task 27, and full-stack API/frontend development). Independent technical decision-making, clear justification of choices, and end-to-end ownership of the project are weighted heavily in the final assessment. All deliverables listed in Day 61 (repository link, daily logs, internship report) are mandatory for submission.